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David J. Watkins, Ph.D.

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Professional Experience

Head of Research **Jun 2026 — Present**
Tutor Intelligence *Watertown, MA*

- Lead robot learning research linking policy training, demonstration quality, and deployment on a 100-robot bimanual fleet
- Led team improvements to full-rate, synchronized data logging and headless teleoperation; co-authored RSS 2026 workshop paper on demonstration quality
- Moved experiment tracking to self-hosted MLflow; designed Rust middleware for data collection and policy execution

Research Lead **Sep 2023 — Jun 2026**
Research Scientist **Dec 2022 — Aug 2023**
RAI Institute (formerly Boston Dynamics AI Institute) *Cambridge, MA*

- Built Foundation Models (10+ researchers) and Capture (30 people) teams
- Led co-design of handheld and robotic grippers for force-based data collection and learned manipulation (Koala, 2026)
- Developed video prediction and multimodal models, and shared policy execution infrastructure used across research teams
- Established research partnerships with Google, Columbia, ETH Zurich, and Agile Robots

Research Fellow **Sep 2018 — Jul 2022**
Army Research Lab *Aberdeen, MD*

- Won 1st place MineRL BASALT Competition (NeurIPS 2021): combined imitation learning, human preference data, and hierarchical knowledge engineering to solve tasks defined only by natural language
- Developed mobile manipulation approaches without runtime localization (IROS 2022 Best Paper Finalist)
- Developed multimodal geometric learning fusing vision and tactile sensing for robotic grasping (ICRA 2019)

Selected Research

BRIDGE: Contact-Rich Manipulation **2026**
RAI Institute, under review at ICRA 2027 *Cambridge, MA*

- Co-authored state-gated diffusion policy experts that combine handheld demonstrations with targeted teleoperation
- Improved task success over handheld-only baselines across three contact-rich manipulation tasks

Koala: Gripper and Data-Capture Co-Design **2026**
RAI Institute, under review at IEEE T-RO *Cambridge, MA*

- Led co-design of handheld and actuated grippers with matched geometry, sensing, and force transmission
- Integrated the platform with data collection and learned policy execution for dexterous manipulation

DF1: Demonstration Quality at Fleet Scale **2026**
Tutor Intelligence, RSS 2026 Workshop: It's the Demos *Watertown, MA*

- Co-authored a demonstration-quality pipeline for a 100-robot bimanual fleet
- Connected teleoperation, data curation, and policy training with human interventions to correct rollout failures

Selected Publications

1. Hajj-Ahmad et al., **Watkins**. "Koala Gripper: Co-designing Robotic Grippers and Data-Capture Devices for Scaling Dexterous Manipulation Learning." *IEEE Transactions on Robotics (T-RO), under review*
2. Surendran, Peri, **Watkins**. "Bridging Handheld and Teleoperated Supervision for Contact-Rich Manipulation via State-Gated Experts." *ICRA 2027, under review*
3. Kapoor et al., **Watkins**. "Maintaining Demonstration Quality in a 100-Robot Teleoperation Pipeline." *RSS 2026 Workshop: It's the Demos*
4. Shang et al., **Watkins**, Herlant. "Theia: Distilling Diverse Vision Foundation Models for Robot Learning." *CoRL 2024*

Education

Ph.D. in Computer Science, *Columbia University* 2017 — 2022
Thesis: *Learning Mobile Manipulation* | Advisor: Prof. Peter Allen | Army Research Lab Fellow
B.S./M.S. in Computer Science, *Columbia University*, 4.0 GPA (M.S.) 2012 — 2017

Skills

ML / Robotics Foundation Models, Diffusion Policies, Imitation Learning, RLHF, Manipulation
Systems Python, C++, Rust, PyTorch, CUDA, ROS 2, Distributed Training
Communication English, Spanish